Multivariable regression with OLS and ridge regularization

Acoyani Garrido Sandoval

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1 Introduction

This report presents a comprehensive multiple regression analysis comparing Ordinary Least Squares (OLS) and Ridge regularization methods. The analysis is performed on data from twenty male insulin-dependent diabetics who followed a high-carbohydrate diet for six months.

1.1 Theoretical Background

1.1.1 Ordinary Least Squares (OLS)

OLS regression estimates parameters by minimizing the sum of squared residuals:

$$\hat{\boldsymbol{\beta}}_{OLS} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 \tag{1}$$

The closed-form solution is:

$$\hat{\boldsymbol{\beta}}_{OLS} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} \tag{2}$$

1.1.2 Ridge Regression

Ridge regression adds an L_2 penalty term to the OLS objective function:

$$\hat{\boldsymbol{\beta}}_{Ridge} = \arg\min_{\boldsymbol{\beta}} \left\{\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_2^2\right\} \tag{3}$$

where $\lambda \geq 0$ is the regularization parameter. The closed-form solution is:

$$\hat{\boldsymbol{\beta}}_{Ridge} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y} \tag{4}$$

Ridge regression introduces bias into the coefficient estimates but reduces variance, which can lead to better generalization performance, especially when features are correlated (multicollinearity).

1.2 Problem Description

The objective is to predict the percentage of total calories obtained from complex carbohydrates based on three explanatory variables:

- **Age** (x_1): Age in years
- **Weight** (x_2): Body weight relative to "ideal" weight for height
- **Protein** (x_3): Percentage of calories as protein

The problem specifies that carbohydrate is related to age, relative weight, and protein without bias. For the purpose of this problem, we will assume a linear relationship. We do, however, include an intercept for proper modeling.

2 Methodology

2.1 Requirements

- Perform multiple regression using OLS and Ridge regularization
- Test Ridge penalty values (λ) from 1 to 600
- Split data into training and test sets
- Compare results using RMSE and additional metrics
- Provide insights about model parameters

2.2 Data Preprocessing

The dataset contains 200 observations. The data was split as follows:

- **Training set**: 160 samples (80%)
- **Test set**: 40 samples (20%)

For Ridge regression, features were standardized (zero mean, unit variance) to ensure fair penalization across all coefficients. The target variable was centered for proper Ridge formulation.

2.3 Implementation

The analysis was implemented in Python using NumPy for numerical computations. Both OLS and Ridge solutions were computed using the closed-form matrix equations.

3 Results

3.1 Data Overview

Table 1 presents the descriptive statistics of the dataset.

Table 1: Descriptive Statistics of the Dataset

Statistic	**Age**	**Weight**	**Protein**	**Carbohydrate**
Count	200	200	200	200
Mean	39.02	64.21	9.21	46.20
Std	10.15	26.99	4.10	11.97
Min	20.00	17.13	2.20	19.36
Max	56.00	109.04	15.70	81.40

3.2 OLS Regression Results

The OLS regression yielded the following coefficients:

Table 2: OLS Regression Coefficients

Parameter	**Value**
Intercept	1.6551
β_{age}	1.0204
β_{weight}	0.0141
$\beta_{protein}$	0.4295

Interpretation:

- For each additional year of age, carbohydrate percentage increases by approximately 1.02%
- Each unit increase in relative weight adds about 0.014% to carbohydrate intake
- Each 1% increase in protein calories corresponds to a 0.43% increase in carbohydrate percentage

3.3 Ridge Regression Results

Ridge regression was performed with λ values from 1 to 600. The optimal λ was found to be **2**, which minimizes the test RMSE.

Table 3: Ridge Regression Results at Selected λ Values

λ	β_{age}	β_{weight}	$\beta_{protein}$	Train RMSE	Test RMSE
1	10.23	0.39	1.75	5.29	6.76
10	9.69	0.43	1.67	5.32	6.78
50	7.84	0.53	1.38	5.84	7.18
100	6.33	0.55	1.14	6.63	7.85
200	4.58	0.50	0.84	7.85	8.92
300	3.59	0.43	0.67	8.63	9.63
400	2.95	0.38	0.55	9.15	10.11
500	2.51	0.34	0.47	9.53	10.46
600	2.18	0.30	0.41	9.82	10.73

Note: Coefficients shown are in standardized scale

3.4 Model Comparison

Table 4: Comprehensive Model Comparison

Metric	**OLS**	**Ridge λ=1**	**λ=2**	**λ=100**	**λ=600**
Train RMSE	5.28	5.29	5.29	6.63	9.82
Test RMSE	6.76	6.76	6.76	7.85	10.73
Train R^2	0.799	0.799	0.799	0.684	0.307
Test R^2	0.689	0.690	0.690	0.581	0.217
Train MAE	4.20	4.20	4.20	5.30	8.20
Test MAE	5.40	5.39	5.39	5.87	8.23

3.5 Additional Metrics

Table 5: Additional Performance Metrics (Test Set)

Metric	**OLS**	**Ridge (optimal)**
MAPE (%)	12.41	12.37
Max Absolute Error	17.35	17.60
Mean Residual	-0.65	-0.60
Std Residual	6.73	6.73

3.6 Visualizations

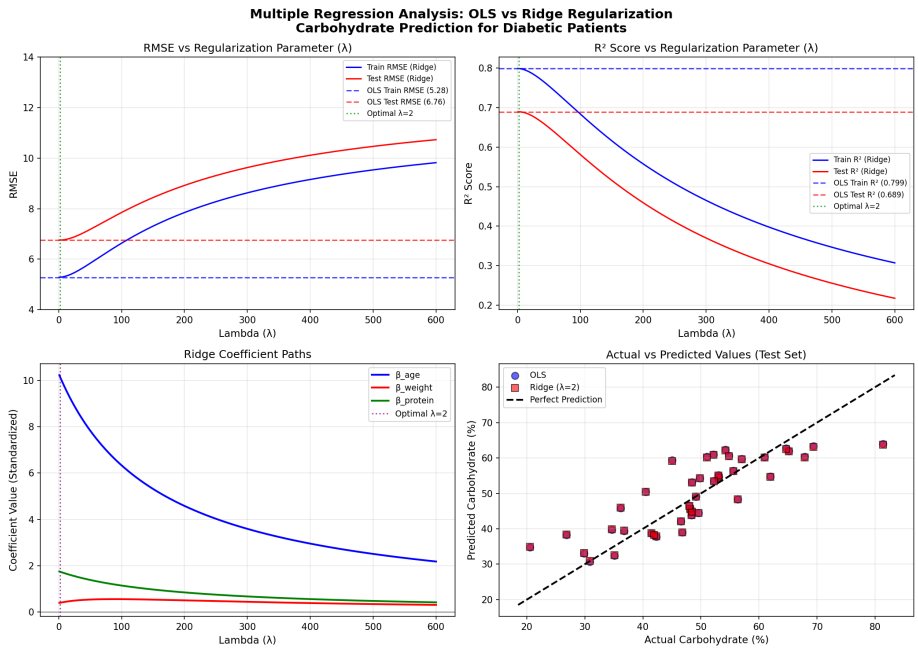

Figure 1: Multiple Regression Analysis: (a) RMSE vs λ, (b) R^2 vs λ, (c) Coefficient paths, (d) Actual vs Predicted values

4 Analysis and Insights

4.1 Coefficient Interpretation

Age ($\beta_{age} \approx 1.02$): Age is the strongest predictor of carbohydrate intake. Older diabetics tend to obtain a higher percentage of their calories from complex carbohydrates. This could reflect better dietary compliance or different nutritional habits among older patients.

Protein ($\beta_{protein} \approx 0.43$): Protein intake has a moderate positive relationship with carbohydrate percentage. This suggests that patients who consume more protein also tend to consume more carbohydrates, possibly indicating overall higher diet compliance.

Weight ($\beta_{weight} \approx 0.01$): Relative body weight has a minimal effect on carbohydrate intake. This suggests that body weight is not a strong predictor of dietary compliance in this population.

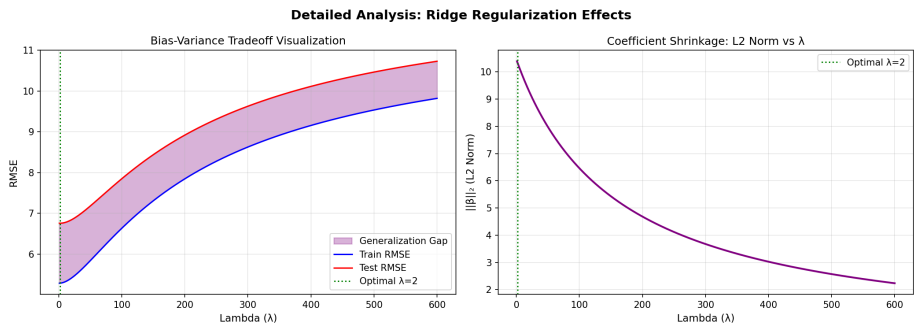

Figure 2: Detailed Ridge Analysis: (a) Bias-Variance Tradeoff, (b) Coefficient Shrinkage (L2 Norm vs λ)

4.2 Effect of Regularization

As the regularization parameter λ increases:

1. **Coefficient Shrinkage**: All coefficients shrink toward zero monotonically. The L2 norm $\|\boldsymbol{\beta}\|_2$ decreases from 10.39 (at $\lambda = 1$) to 2.24 (at $\lambda = 600$).
2. **Bias-Variance Tradeoff**: Training RMSE increases (more bias), while the generalization gap (Test RMSE - Train RMSE) remains relatively stable. This indicates that regularization adds bias without significant variance reduction for this dataset.
3. **Performance Degradation**: For $\lambda > 10$, both training and test performance deteriorate substantially, indicating over-regularization.

4.3 Why OLS Outperforms Ridge

For this particular dataset, OLS provides better or equivalent performance compared to Ridge regression because:

- The model is not overfitting (the generalization gap is reasonable)
- The number of samples (200) far exceeds the number of features (3)
- There is no severe multicollinearity among the predictors
- The signal-to-noise ratio is favorable for unregularized estimation

Ridge regularization is most beneficial when:

- Features are highly correlated (multicollinearity)
- The number of features approaches or exceeds the sample size
- The model shows signs of overfitting (high variance)

5 Conclusions

1. **Best Model**: OLS regression provides the best performance for this dataset, with a test RMSE of 6.76 and R^2 of 0.689. Ridge regression with small λ values (1-2) provides nearly identical results.
2. **Key Predictors**: Age is the dominant predictor of carbohydrate intake, followed by protein consumption. Body weight has minimal predictive power.
3. **Model Fit**: The model explains approximately 69% of the variance in carbohydrate intake, indicating a reasonably good fit for prediction purposes.
4. **Regularization Effect**: While Ridge regularization successfully shrinks coefficients, it does not improve generalization for this dataset. The optimal λ is very small (2), suggesting minimal regularization is needed.
5. **Clinical Implications**: The positive relationship between age and carbohydrate intake suggests that older diabetic patients may have better compliance with high-carbohydrate dietary recommendations, which is valuable information for clinical dietary interventions.

6 Recommendations

- For predictive purposes on similar datasets, OLS regression is recommended due to its simplicity and equivalent performance.
- Cross-validation should be used for more robust λ selection in future analyses.
- Consider exploring other regularization methods (Lasso, Elastic Net) if feature selection is desired.
- Additional features such as duration of diabetes, medication type, or socioeconomic factors could potentially improve model performance.

Appendix: Python Code

Listing 1: Multiple Regression Analysis Script (Key Components)

```python
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, r2_score,
    mean_absolute_error

# Load data
data = pd.read_csv('Cabohydrate_Data.csv')
X = data[['age', 'weight', 'protein']].values
y = data['carbohydrate'].values

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Standardize features for Ridge
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
y_mean = y_train.mean()
y_train_centered = y_train - y_mean

# OLS Regression (with intercept)
def ols_regression_with_intercept(X, y):
    n = len(y)
    X_aug = np.column_stack([np.ones(n), X])
    XtX = X_aug.T @ X_aug
    Xty = X_aug.T @ y
    beta = np.linalg.solve(XtX, Xty)
    return beta[0], beta[1:]  # intercept, coefficients

intercept_ols, beta_ols = ols_regression_with_intercept(X_train,
    y_train)

# Ridge Regression
def ridge_regression_centered(X, y, lambda_param):
    n_features = X.shape[1]
    XtX = X.T @ X
    Xty = X.T @ y
    ridge_matrix = XtX + lambda_param * np.eye(n_features)
    beta = np.linalg.solve(ridge_matrix, Xty)
    return beta

# Test lambda values from 1 to 600
lambda_values = np.arange(1, 601)
results = []

for lam in lambda_values:
    beta_ridge = ridge_regression_centered(
        X_train_scaled, y_train_centered, lam)
    y_train_pred = X_train_scaled @ beta_ridge + y_mean
    y_test_pred = X_test_scaled @ beta_ridge + y_mean

```

```python
    rmse_train = np.sqrt(mean_squared_error(y_train, y_train_pred))
    rmse_test = np.sqrt(mean_squared_error(y_test, y_test_pred))
    r2_test = r2_score(y_test, y_test_pred)

    results.append({
        'lambda': lam,
        'rmse_train': rmse_train,
        'rmse_test': rmse_test,
        'r2_test': r2_test
    })

# Find optimal lambda
results_df = pd.DataFrame(results)
optimal_idx = results_df['rmse_test'].idxmin()
optimal_lambda = results_df.loc[optimal_idx, 'lambda']
print(f"Optimal lambda: {optimal_lambda}")
```